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Dirty Spokes Dirtbike Rental Management System

Step 3 Draft

Web App URL: http://classwork.engr.oregonstate.edu:41801/

a) Project Outline, DB Outline, ERD Schema, & Sample Data Updated Version:

Overview

Dirty Spokes is a recreational dirtbike rental facility and track located in sunny Southern California, operating a fleet of 50 bikes across 8 major brands including Honda, Yamaha, and Kawasaki. The facility services roughly 500 unique customers per year across an estimated 2,500 annual rental sessions. Currently, staff manage bike inventory, customer records, and rental history through manual bookkeeping and paper waivers.

Over the past year, this has led to a number of operational issues including customer records being lost or duplicated, no reliable way to track when a customer had which bike at any given time, and difficulty identifying when a bike was last rented or how many hours it has accumulated.

A database-driven web application will allow Dirty Spokes administrators to manage Bikes, Brands, Customers, and Rentals in a unified system, ensuring accurate rental records, up-to-date usage metrics for bike maintenance and popularity, and smoother day-to-day operations at the track.

In applying the normalization process to this design, no changes to the outline or ERD are required after Step 1's feedback fixes. The schema is in 3NF. All attributes contain atomic values with no repeating groups, all non-key attributes are fully dependent on their primary keys with no partial dependencies, and no non-key attribute determines another non-key attribute.

Foreign key ON DELETE and ON UPDATE behaviors have been documented in Step 3 for clarity on relationships in the database outline.

Database Outline

Bikes: records the dirtbikes available in the rental fleet
- ○ bikeID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- ○ frameNumber: VARCHAR(50), unique, NOT NULL
- ○ brandID: INT UNSIGNED, NOT NULL, FK → Brands.brandID
- ○ modelName: VARCHAR(100), NOT NULL
- ○ engineSize: VARCHAR(10), NOT NULL - e.g. "250cc", "450cc"
- ○ bikeYear: INT UNSIGNED, NOT NULL
- ○ engineHourMeter: DECIMAL(6,1), NOT NULL
- ○ Relationships:
    - ■ M:1 between Bikes and Brands, many bikes can belong to one brand implemented by FK brandID. ON DELETE RESTRICT prevents a brand from being deleted while bikes are associated with it. ON UPDATE CASCADE propagates brand ID changes to all associated bikes.
    - ■ 1:M between Bikes and Rentals, one bike can appear in many rental records implemented by FK bikeID in Rentals

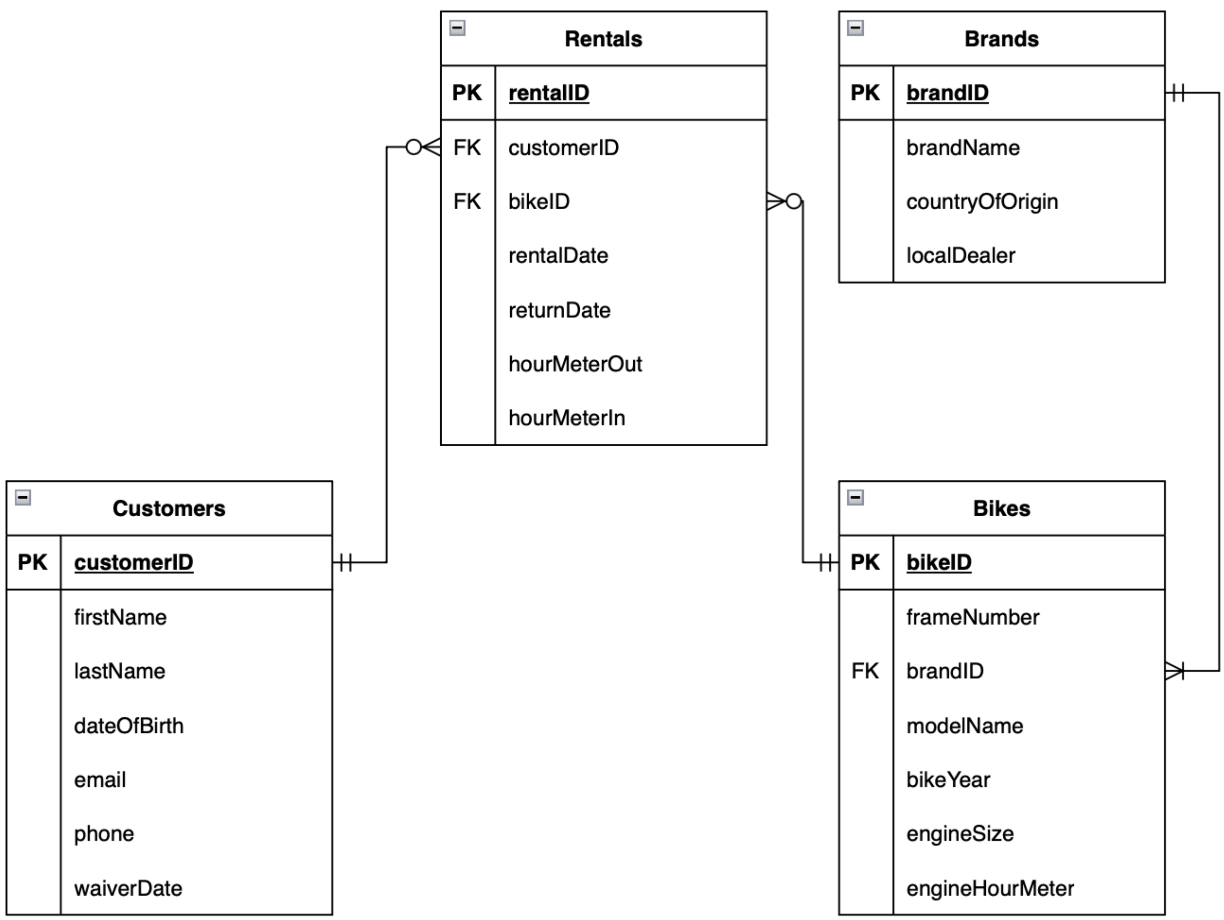
Brands: records the manufacturers of the bikes in the fleet
- ○ brandID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- ○ brandName: VARCHAR(100), unique, NOT NULL
- ○ countryOfOrigin: VARCHAR(100), NOT NULL
- ○ localDealer: VARCHAR(255), NOT NULL
- ○ Relationships:
    - ■ 1:M between Brands and Bikes, one brand can have many bikes implemented by brandID FK in Bikes

Customers: records customers who rent bikes at the track
- ○ customerID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- ○ firstName: VARCHAR(50), NOT NULL
- ○ lastName: VARCHAR(50), NOT NULL
- ○ dateOfBirth: date, NOT NULL
- ○ email: VARCHAR(255), unique, NOT NULL
- ○ phone: VARCHAR(20), NOT NULL
- ○ waiverDate: date, NOT NULL
- ○ Relationships:
    - ■ 1:M between Customers and Rentals, one customer can have many rental records implemented by FK customerID in Rentals

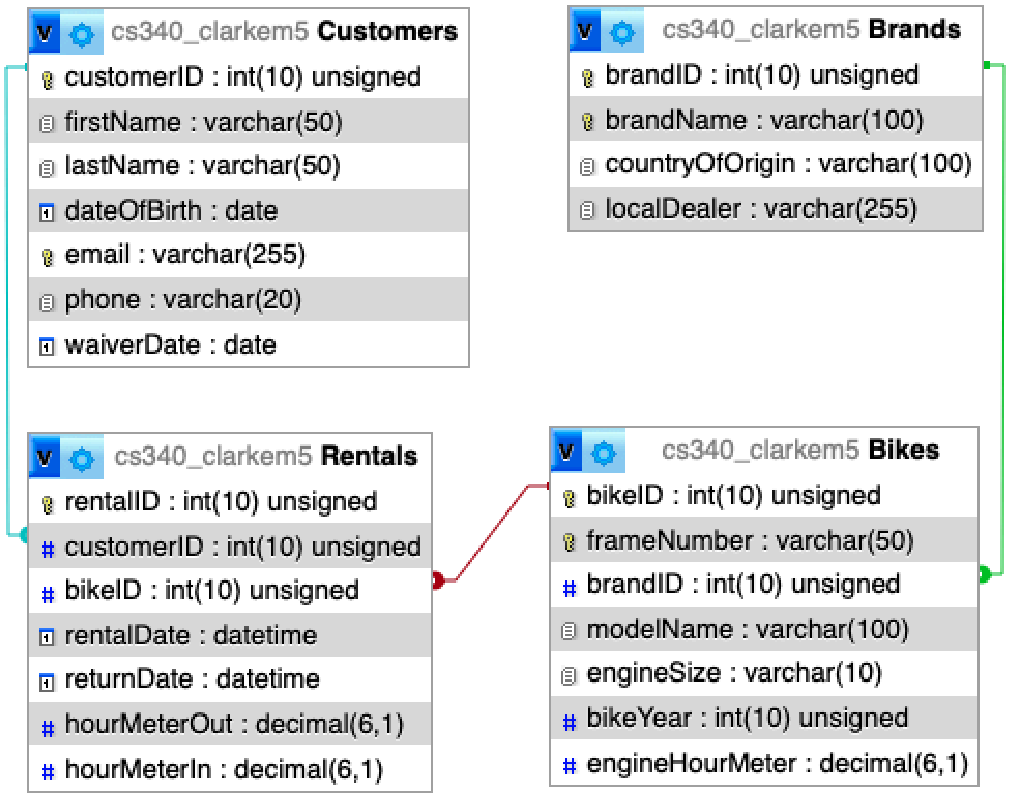
Rentals: intersection table recording which customer rented which bike and when
- ○ rentalID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- ○ customerID: INT UNSIGNED, NOT NULL, FK → Customers.customerID
- ○ bikeID: INT UNSIGNED, NOT NULL, FK → Bikes.bikeID
- ○ rentalDate: datetime, NOT NULL

- returnDate: datetime, NOT NULL
- hourMeterOut: DECIMAL(6,1), NOT NULL
- hourMeterIn: DECIMAL(6,1), NOT NULL
- Relationships:
  - M:1 between Rentals and Customers, many rental records can belong to one customer where Customers.customerID is PK and Rentals.customerID is FK. ON DELETE RESTRICT prevents deletion of a customer with existing rental records. ON UPDATE CASCADE propagates ID changes.
  - M:1 between Rentals and Bikes, many rental records can reference one bike where Bikes.bikeID is PK and Rentals.bikeID is FK.
    ON DELETE RESTRICT prevents deletion of a bike with existing rental records. ON UPDATE CASCADE propagates ID changes.
  - M:N between Customers and Bikes, a customer can rent many bikes and a bike can be rented by many customers

Entity-Relationship Diagram



Schema Diagram



Example Data

Bikes Table

bikeID	frameNumber	brandID	modelName	engineSize	bikeYear	engineHourMeter
1	JYACB11C0RA015607	3	YZ65	65cc	2024	15.8
2	VBKXWM236MM321033	2	300 XC-W	293cc	2021	71.4
3	VNBS648C4PB000307	1	SE-F 500 Factory 4T	478cc	2023	225.5

Brands Table

brandID	brandName	countryOfOrigin	localDealer
1	Sherco	France	Trials Offroad
2	KTM	Austria	SoCal Moto
3	Yamaha	Japan	SoCal Moto

Customers Table

customerID	firstName	lastName	dateOfBirth	email	phone	waiverDate
1	Jeremy	Stevens	1989-06-11	jsteve0@email.com	555-0101	2024-11-22
2	Andrea	Hernandez	2003-11-25	andreah44@email.com	555-0202	2025-04-17
3	Roman	Lutoslawski	1978-08-26	roman444@email.com	555-0303	2020-08-20

Rentals Table

rentalID	customerID	bikeID	rentalDate	returnDate	hourMeterOut	hourMeterIn
1	3	3	2025-04-06 11:22:56	2025-04-08 16:43:16	219.9	225.5
2	3	2	2026-04-10 09:30:17	2026-04-13 10:17:44	60.6	68.0
3	2	2	2026-04-19 10:19:19	2026-04-19 15:22:04	68.1	71.4

b) Fixes based on Feedback from Previous Steps:

Step 1 Accepted Fixes

- Add lengths to VARCHAR and INT attributes - VARCHAR lengths, INT signs, and DECIMAL placing have been added to all attributes throughout the Database Outline for greater precision and clarity.
- Make Brand mandatory to Bikes - The Brands→Bikes relationship in the ERD has been updated to reflect a mandatory relationship, as every bike in the fleet must be associated with a brand.
- Remove M:N connector between Customers and Bikes in ERD - The direct line between Customers and Bikes has been removed from the ERD. This relationship is already properly represented through the Rentals intersection table and the direct connector was redundant.
- Add cardinality to connections in the ERD - Proper cardinality markers have been added to all relationship lines in the ERD.
- Edit engineHours to singular - engineHours was not present in the Database Outline so it has been added to the Bikes entity and changed to a singular name, engineHourMeter.

Step 1 Declined Suggestions

- Add Employees table - The scope of this system is intentionally limited to bike inventory, customer records, and rental history. This is an administrator-facing database tool, not an employee management system. Adding an Employee table would exceed the intended scope of the project.
- phone attribute changed from VARCHAR to INT - We chose to keep phone as VARCHAR since phone numbers may include formatting characters such as dashes, parentheses, spaces, and country code prefixes. Storing phone as INT would risk data loss and overflow on numbers with country codes as well as prevent storing any non-numberic character formatting.

- Make localDealer optional - This suggestion was not employed as recording the nearest outside facility that sells or services a specific brand is a necessity in assembling a fleet of popular models and maintaining them throughout their lifecycle. The scenario of a major brand ceasing operations and no longer being available is highly unlikely, and the discontinuation of a brand would result in its removal from the active fleet entirely rather than keeping it with a null dealer field.
- Add license and insurance attributes to Customers - This suggestion is not a necessity in this context. 50cc MX bikes and increasingly popular electric bikes of comparable sizes are geared toward riders as young as 5 years old who would not have a license or insurance to record. Beyond this, off-highway vehicles do not require a license or insurance to operate in legal settings such as a designated MX or enduro track. Adding these fields would therefore not be applicable to a significant portion of the customer base and is out of scope for the current system design.

Step 2 Feedback

<u>Review by Ikumi Kameyama:</u>

1. Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?
Overall, the schema follows the outline and ERD very closely. The four entities (Bikes, Brands, Customers, and Rentals), match across the outline, ERD, and schema.
A few small observations:

- The ERD includes all attributes now, which matches the outline well.
- The schema diagram is clear, but the ERD uses “bike Year” (with a space) while the schema uses bikeYear. This is a small formatting inconsistency.
- The ERD shows the correct cardinalities, and the schema enforces them with foreign keys.

2. Is there consistency in naming (overview, outline, ER, schema), plural entities, singular attributes, capitalization)?

- Entities use PascalCase (Bikes, Brands, Customers, Rentals).
- Attributes use camelCase or lowercase with no spaces (frameNumber, engineHourMeter, bikeYear).

3. Is the schema easy to read (diagram clear, lines not crossed)?
Yes, the ERD is clean and easy to read. The cardinality markers are clear, and the relationships are not cluttered. The schema diagram is also readable and well-organized.
4. Are intersection tables properly formed (two FKs and facilitate a M:N relationship)?
Yes. The Rentals table correctly acts as the intersection table for the M:N relationship between Customers and Bikes.

- It includes customerID (FK)
- It includes bikeID (FK)
- It includes its own PK (rentalID), which is acceptable
- It includes additional attributes relevant to the rental event

5. Does the sample data suggest any non-normalized issues (partial dependencies or transitive dependencies)?
I did not find any normalization issues.
6. Is the SQL file syntactically correct?
The SQL shown in the schema diagram appears correct.

The example data loads successfully in MariaDB, which suggests the SQL is valid.
7. Are the data types appropriate considering the description of the attribute in the database outline?

- DECIMAL(6,1) for hour meters is correct
- VARCHAR lengths are reasonable
- INT UNSIGNED for IDs is appropriate
- DATETIME for rentalDate and returnDate is correct

8. Are primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?
Yes, the primary and foreign keys match the outline and ERD.
The schema enforces the correct relationships.
9. Are relationship tables present when compared to the ERD/Schema?
Yes.
The Rentals table is present and correctly implements the M:N relationship between Customers and Bikes.
10. Is all example data shown in the PDF inserted?
Yes.
The example data in the PDF matches the schema and demonstrates the relationships clearly.
11. Is the SQL well-structured and commented (hand-authored vs. exported)?
The SQL appears hand-authored and well organized.
12. Originality statement
All feedback provided here is my own work based on my understanding of the project. I did not use AI tools to generate this review.

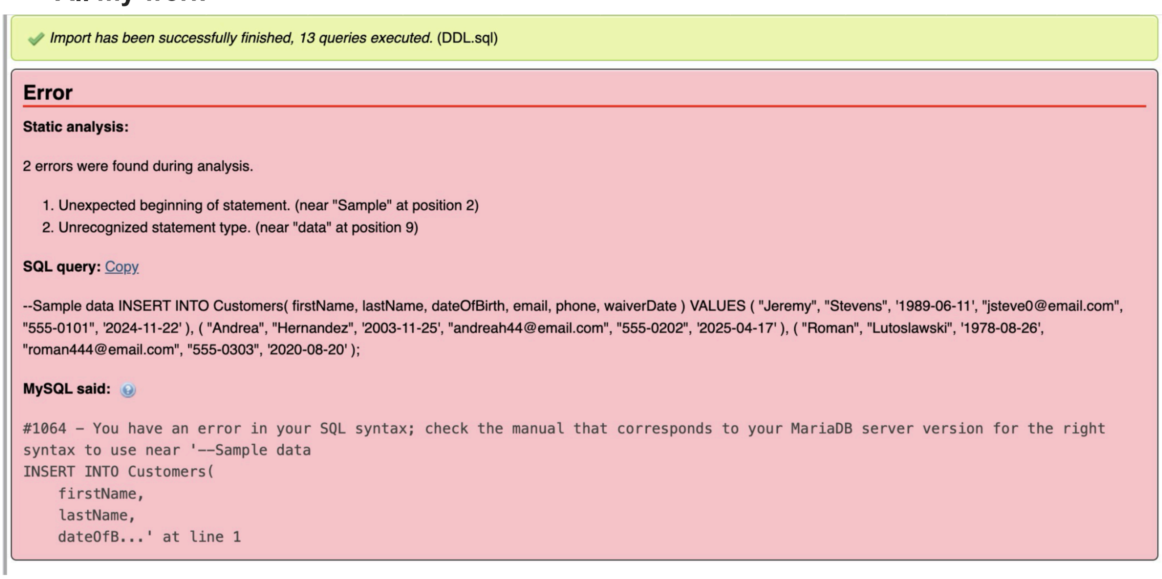
Review by Hannah Kemplin:

Hi Spencer and Matthew!
You all have done some really solid work with your Dirty Spokes rental facility concept. It looks like your project is off to a great start!

- *Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?*
  Yes, the schema accurately models the database outline and ERD. The diagram correctly depicts all proposed tables and attributes as well as their respective relationships. The only (very minor) inconsistency I detected was that the bikeYear and engineSize attributes are switched in order on the ER and schema diagrams.
- *Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?*
  Naming is consistent across overview, outline, ERD, and schema, utilizing plural PascalCase for the entities and singular camelCase for the attributes.
- *Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?*
  The schema is clear and readable. The layout of the tables is neat, making the relationship lines clean and easy to follow.
- *Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?*
  The Rentals intersection table is properly formed, featuring foreign keys customerID from Customers and bikeID from Bikes and facilitating a M:N relationship between Customers and Bikes.
- *Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?*

The database design seems to be normalized, achieving 3NF. I could not find any further non-normalized issues.

- *Is the SQL file syntactically correct? This can be easily verified by using PhpMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)*
  The SQL file was successfully imported into my CS340 database using PhPMyAdmin. I did, however, receive an error message indicating a syntax error (screenshot included at the end of my review).
- *In the SQL, are the data types appropriate considering the description of the attribute in the database outline?*
  The data types in the SQL are appropriate given the description of the attribute in the database outline.
- *In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?*
  Primary keys and foreign keys for all entities are correctly defined, and referential integrity is preserved by the CASCADE operations declared in the creation of the Rentals table.
- *In the SQL, are relationship tables present when compared to the ERD/Schema?*
  All tables, including Bikes, Customers, Rentals, and Brands, are present in the SQL when compared to the ERD and schema.
- *In the SQL, is all example data shown in the PDF INSERTED?*
  Yes, all example data shown in the PDF is properly inserted in the SQL.
- *Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?*
  Overall, the SQL is well-structured. It is commented and appears to be hand authored.
- *As a reviewer, clearly describe to what extent your feedback to the team was original (e.g. "all my work") or non-original (e.g. used AI tools per Code Citation Tips).*
  All my work



<u>Review by Kayla Haney:</u>

Hi Matthew and Spencer,
This is a well-structured database design with a matching schema and clean SQL. The relationships and entities remain consistent throughout the DDL, sample data and ERD, and I did not find any syntax errors.

Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?
Yes, the schema matches the ERD and database outline.
Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?
Yes, there is consistent naming throughout the ER and schema entities/attributes with the appropriate capitalization.
Is the schema easy to read?
The schema is easy to follow with no relationships crossed.
Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)? Does the sample data suggest any non-normalized issues?
The four tables (Customers, Brands, Rentals and Bikes) are clear and have understandable relationships. There is a clear M:N relationship between Customers and Bikes with Rentals as the intersection table for recording which customers rented which bike. The schema is in 3NF with no partial dependencies or transitive dependencies.
The SQL file:
The SQL file is syntactically correct, and the datatypes are appropriate according to the database outline descriptions. The PK and FKs are correctly defined and the CASCADE operations are appropriately declared. These match the schema, ER and database outline.
The sample data is inserted with data that follows the schema. All the example data in the pdf is presented and INSERTED correctly. Overall, the SQL is well-structured with clean, descriptive comments. Great work on the SQL and schema.
This review was all written by me.

Review by Sylvia Nguyen:

Hi Spencer and Matthew!
Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?
Yes the schema presents a physical model that follows the database outline and ERD.
Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?
Yes there is consistency in the naming between overview, outline, ERD and schema. The entities are plural and capitalized and the attributes adhere to camelcase.
Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?
The schema is easy to read and clear. Relationship lines are not crossed.
Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?
Yes the intersection tables are properly formed with two FKs to faciliate the M:N relationship.
Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?
The sample data appears to satisfy 3NF, where each table has a primary key and there are no composite keys. There are no partial or transitive dependencies.
In the SQL, are the data types appropriate considering the description of the attribute in the database outline?
Yes the data types are appropriate.
In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?

Yes the primary and foreign keys are correctly defined comapred to the schema. Yes there are appropropriate cascade operations.
In the SQL, are relationship tables present when compared to the ERD/Schema?
Yes.
In the SQL, is all example data shown in the PDF INSERTED?
Yes.
Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?
Yes comments are clear to introduce the project and act as labels for the tables as well.
As a reviewer, clearly describe to what extent your feedback to the team was original (e.g. "all my work") or non-original (e.g. used AI tools per Code Citation Tips).
My review is original.

Step 2 Accepted Fixes

All four peer reviews found no critical issues with the schema, ERD, DDL, or sample data. No required fixes were identified across the feedback.

Step 2 Declined Suggestions

- bikeYear spacing inconsistency - one reviewer noted a space in “bike Year” in the ERD, however upon review there is no such spacing error present. The ERD correctly displays “bikeYear” consistent with the schema and database outline. No change was made.
- Attribute order of bikeYear and engineSize in the ERD and schema diagrams - one reviewer noted these attributes appear in a different order between the two diagrams. The ordering difference does not affect functionality, correctness, or readability and was not changed.
- phpMyAdmin SQL import error - one reviewer noted a syntax error when importing the DDL via phpMyAdmin. This was determined to be a phpMyAdmin issue with “--” comment parsing and not a true SQL syntax error. The DDL executes without errors on ENGR MariaDB and no changes to the SQL were made.

Citations

The web application scaffolding for Step 3 was written with the help of Claude (Anthropic). Prompts described our database schema and requested UI scaffolding templates for a Node.js/Express/Handlebars application.

All other work in this document was done without the use of AI tools.